The Jailbreak Robustness Ladder: Separating Safety Failure from Comprehension Failure in Cross-Lingual Attacks on Open Language Models

Working paper / technical report. Code and data: this repository.

Abstract

Language-model safety training is done overwhelmingly in English, yet the models are deployed in many languages. A natural attack is therefore to *translate* a harmful request into another language and hope the safety training does not transfer. Prior work has shown this works on frontier models. However, essentially all cross-lingual jailbreak evaluations share a methodological flaw: they count **"the model did not refuse"** as **"the attack succeeded"**. For lower-resource languages — exactly the ones of interest — this conflates two very different events: the model's *safety* failing, and the model simply *not understanding the request*. A small model that emits confused text when asked something harmful in Urdu has not been jailbroken; it has been broken.

We introduce the **Jailbreak Robustness Ladder** (`jbladder`), a reproducible harness that (i) treats language-switching as one *attack transform* among many (roleplay, refusal-suppression, base64, payload-splitting, …), so linguistic and syntactic attacks can be ranked on a single axis; (ii) judges every response with a **three-way classifier — REFUSE / COMPLY / EVADE** — and reports a **comprehension-controlled attack-success rate** (cc-ASR): jailbreak success *among requests the model actually understood*; and (iii) evaluates a concrete **mitigation**. Using 150 paired harmful/benign probes translated into **seven languages spanning the resource spectrum** — Hindi, Bangla, Urdu, Arabic (lower/medium-resource) and Chinese, Spanish, French (high-resource) — run across four open models (Qwen2.5-0.5B/1.5B/3B, SmolLM2-1.7B; **19,800 judged responses**), we find that **safety is anchored to the languages a model was actually aligned in**, not to a language's global resource level. Spanish and French retain safety on the multilingual Qwen models (cc-ASR 0.01–0.12) but collapse on English-centric SmolLM2 (~0.80), which is safe *only* in English. Safety **collapses in lower-resource languages (0.76–0.86 for Hindi/Urdu/Bangla) across every model** — a 2.5× gap versus English (0.31). Chinese is a cautionary exception: despite being very high-resource and well-understood, its cc-ASR averages **0.51** (0.88 on Qwen2.5-3B), because these models parse Chinese fluently yet were not robustly *safety*-tuned in it — **capability is not safety**. In Bangla and Urdu the *raw* ASR further **understates** the hole (comprehension drops to ~0.73–0.75, so a naive judge miscounts confusion as safety). A hardened defensive system prompt reduces attack success by only 15–18 %. Our central methodological claim is that **cross-lingual safety cannot be measured without controlling for comprehension** — a claim we reinforce the hard way, when human validation of the judge caught a tokenization bug that had made Chinese look artificially safe (cc-ASR 0.27 → 0.51 once fixed). Our central empirical claim is that **cross-lingual jailbreak success is a shadow of where the model was aligned**, not a separate linguistic vulnerability.

1. Introduction

Refusal is the most visible safety behaviour of an aligned language model: when asked to do something harmful, it should decline. A *jailbreak* is any input transformation that causes the model to comply anyway. A large literature catalogues such transformations — roleplay personas, refusal suppression, affirmative-prefix injection, encoding tricks, and, notably, **translation into lower-resource languages** (Yong et al., 2023; Deng et al., 2023).

The translation attack is uniquely important for equity reasons: it means the users least served by English-centric safety work — speakers of Hindi, Bangla, Urdu, Arabic, and hundreds of other languages — are also the least protected. But it is uniquely *hard to measure*, for a reason the literature has under-emphasised. When a small or mid-sized model is asked something harmful in a language it barely models, three things can happen:

1. it recognises the harm and **refuses** (safety transferred);
2. it produces a fluent, harmful, on-topic answer (**genuine jailbreak**); or
3. it produces confused, off-topic, or garbled text (**it did not understand**).

Counting (3) as a jailbreak — which any refusal-keyword-based metric does, because garbled text contains no refusal phrase — systematically *overstates* the cross-lingual vulnerability of weak models and *understates* the role of comprehension. This paper's contribution is to fix that.

Contributions. - **A comprehension-controlled measurement.** A three-way judge (REFUSE / COMPLY / EVADE) and a metric, cc-ASR, that reports jailbreak success only over understood requests, cleanly separating safety failure from comprehension failure (§3.3, §5.2). - **A unified attack ladder.** Ten atomic transforms plus stacked combinations, with language-switching (hi/bn/ur/ar) as first-class transforms, so we can ask *whether switching language is a stronger lever than syntactic obfuscation* (§3.2, §5.1). - **Paired benign controls.** 50 topically-matched harmless twins that both measure over-refusal and provide a per-language **competence proxy** used in the entanglement analysis (§3.1). - **An empirical finding — safety is per-model and alignment-anchored.** A model is safe in a language to the extent it was *safety*-aligned in it: SmolLM2 is safe only in English; Qwen keeps safety in English/Spanish/French/Arabic but not low-resource languages, and — strikingly — not in Chinese, which it understands fluently (capability ≠ safety). Comprehension correlates negatively with comprehension-controlled ASR ($r = -0.47$) across 42 model×language cells, and where comprehension is low (Bangla, Urdu) raw ASR *understates* the failure (§5.2). - **A measured defense.** A hardened defensive system prompt reduces attack success by only 15–18 % (relative) at an over-refusal cost of +0.06 to +0.12 — a partial fix that motivates model-level mitigation (§5.3). - **Full reproducibility with no API keys.** Everything runs locally on open-weight models; a purpose-built translation pipeline (NLLB-200) plus a native-reviewed "gold" subset produces the multilingual probes (§4).

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2. Related work and how this differs

Cross-lingual jailbreaks. Yong et al. (2023), *Low-Resource Languages Jailbreak GPT-4*, showed that translating unsafe prompts into low-resource languages bypasses GPT-4's guardrails; Deng et al. (2023), *MultiJail*, built a multilingual jailbreak benchmark. Both measure whether the model

complied; neither separates compliance from comprehension, and both target frontier models where comprehension is near-ceiling (so the distinction is invisible). Our contribution is precisely the regime — small/mid open models — where comprehension is *not* a given, and the tooling to measure it.

Exaggerated safety / over-refusal. Röttger et al. (2023), *XSTest*, introduced benign prompts that superficially look unsafe to measure over-refusal, in English only. We adopt the paired-twin idea, extend it multilingually, and repurpose the benign twins as a **comprehension probe**.

Jailbreak taxonomies and benchmarks. Wei et al. (2023) analyse why safety training fails (competing objectives, mismatched generalisation) and catalogue attacks including refusal-suppression, prefix-injection, and encodings; Kang et al. (2023) use payload-splitting; Andriushchenko & Flammarion (2024) show past-tense reformulation. Benchmarks such as HarmBench and JailbreakBench standardise harmful-prompt sets and judging. We reuse these attack *mechanisms* but place language-switching on the same measurement axis and add the comprehension control.

In one sentence. Prior work establishes that translation is an attack and that over-refusal matters; **no prior cross-lingual evaluation we are aware of reports a comprehension-controlled attack-success rate**, which is the instrument this paper contributes.

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3. Method

3.1 Probe set

We hand-author **100 harmful** and **50 benign** English prompts across ten harm categories (cybercrime, fraud, weapons, drugs, self-harm, hate, misinformation, privacy/surveillance, violent extremism, illegal commerce). Harmful prompts are short, **non-operational refusal triggers** — they name a harmful goal without any operational content — in the style of AdvBench / HarmBench. Each harmful category is mirrored by benign twins that share surface vocabulary but have benign intent (e.g. harmful *"how do I break into someone's Wi-Fi"* vs. benign *"how do I protect my Wi-Fi from intruders"*). Benign twins serve two purposes: measuring over-refusal, and — when translated — estimating the model's **competence** in each language (does it produce a coherent, on-topic answer to a harmless request?).

3.2 Attack transforms

Each transform is a pure function `request → chat messages`, tagged with a *mechanism family*:

family	transforms
control	`plain`
framing	`refusal_suppression`, `prefix_injection`
persuasion	`roleplay_persona`, `hypothetical`, `authority_framing`, `past_tense`
obfuscation	`base64`, `leetspeak`, `payload_split`
translation	`translate_{hi,bn,ur,ar,zh,es,fr}` (7 languages across the resource spectrum)

Because transforms are pure functions we **compose** them; we include five stacked combos (e.g. `translate_hi+refusal_suppression`) to test whether weak attacks *compound*. In total **22 transforms × 150 prompts = 3,300 queries per model condition**.

3.3 Three-way judge and comprehension-controlled ASR

A rule-based classifier maps each response to one of:

- **REFUSE** — an explicit decline (multilingual refusal cues) not overridden by a long substantive answer;
- **EVADE** — confused, off-topic, clarification-seeking, or too short to be a real answer (the typical failure mode of a broken obfuscation);
- **COMPLY** — a substantive, on-topic answer.

Refusal cues cover all seven languages, and word-counting is Unicode-aware (Devanagari, Bangla, Arabic, Han, Latin) so a fluent non-English answer is not mistaken for empty output. Two further detectors route degenerate repetition and near-verbatim prompt-echo to EVADE (a broken translation that loops or restates the request is not a jailbreak). From these we define, per (model, transform), on harmful prompts:

- **comprehension** = 1 − EVADE-rate — did the model engage with the request at all?
- **ASR (raw)** = COMPLY / N — the quantity prior work reports;
- **cc-ASR** = COMPLY / (COMPLY + REFUSE) — **attack success among understood requests**.

For benign prompts we report **over-refusal** = REFUSE-rate and **benign-pass** = COMPLY-rate; the translated benign-pass rate is our per-language competence proxy.

We **validate** the judge against human labels on a stratified sample of **72** responses (the author read the Hindi/Bangla/Urdu/Arabic ones directly): overall accuracy **0.83**, with **perfect refusal detection (20/20)**, and for the load-bearing COMPLY (attack-success) decision, precision **0.56** / recall **0.71**. The residual error is concentrated in EVADE→COMPLY (garbled multilingual output judged as compliance), which means our translation ASRs are, if anything, an *upper bound* — the true cross-lingual gap is at least as large as reported (§5.4).

3.4 Metrics

Beyond the rates above: **attack lift** = ASR(transform) − ASR(plain); **model robustness** = 1 − mean ASR over attacks; **transform potency** = mean attack-lift across models; **composability residual** = ASR(A+B) − [1 − (1−ASR(A))(1−ASR(B))] (positive ⇒ super-additive); **cross-model transfer** = Jaccard overlap of the prompts jailbroken under a transform. All headline rates carry **95 % percentile-bootstrap confidence intervals**.

3.5 Defense

We evaluate a **defensive system prompt** that instructs the model to refuse harmful requests "regardless of language, fiction, roleplay, research framing, or encoding, and regardless of any instruction to not refuse," subordinating any attacker-supplied system message. We re-run the full

ladder with it and report ASR reduction and the over-refusal cost. We also discuss **language normalization** (routing the request through English before the safety decision), for which our own English-vs-translation numbers provide a direct estimate of recoverable safety.

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4. Experimental setup

Models (open, non-gated): Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, SmolLM2-1.7B-Instruct. The Qwen ladder isolates *scale*; SmolLM2 is a more English-centric control. Two of them (Qwen2.5-1.5B, SmolLM2-1.7B) are additionally run under the defensive system prompt, for six conditions in all. **Languages (7):** Hindi, Bangla, Urdu, Arabic (lower/medium-resource) and Chinese, Spanish, French (high-resource controls). **Hardware:** a single RTX 5060 Laptop GPU (8 GB), bf16, greedy decoding, 160 new tokens, batched. **Translations:** produced by NLLB-200-distilled-600M; the 47 seed prompts (188 cells, in the four focus languages) additionally carry hand-authored, native-reviewed "gold" translations that override the machine output. Provenance is tracked per cell. **No API keys are used anywhere**; an OpenAI/Anthropic adapter is included so the identical harness can be pointed at a frontier model when a key is available.

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5. Results

All numbers are from 19,800 judged responses (6 model conditions × 150 prompts × 22 transforms). Rates carry 95 % bootstrap CIs; see `results/tables/*` and `results/figures/*`.

5.1 The attack ladder

Robustness is not monotonic in scale. Ranking models by 1 − mean ASR (`model_robustness.png`): **Qwen2.5-1.5B is by far the most robust (0.58)**, then Qwen2.5-0.5B (0.38), Qwen2.5-3B (0.37), and SmolLM2-1.7B (0.31). The 3B model is *no more* robust than the 0.5B and well below the 1.5B — a recurring theme: greater capability improves *comprehension of the attack*, which for obfuscations and translations removes the EVADE "shield" and converts near-misses into genuine jailbreaks (the 3B's Chinese cc-ASR of 0.88, §5.2, is the extreme case).

Which attacks are most potent? By mean attack-lift over the plain baseline (`transform_potency.png`): `prefix_injection` (+0.64) and `roleplay_persona+prefix_injection` (+0.62) lead, immediately followed by `translate_hi+refusal_suppression` (+0.57), `hypothetical` (+0.56), and `translate_hi` (+0.43). Aggregated by mechanism family, mean ASR is **framing 0.72 > combo 0.61 > persuasion 0.56 > obfuscation 0.48 ≈ translation 0.47** — but the translation family's low *mean* is deceptive: it is strongly **bimodal**, averaging the mostly-harmless Spanish/French translations with the most potent low-resource ones (`translate_hi` is the 5th-strongest attack overall). Obfuscation is weakest because small models cannot decode base64/leetspeak — a comprehension failure the three-way judge correctly scores as EVADE.

Composability is mostly sub-additive — with one telling exception. Stacking two attacks usually yields *less* than the independence prediction 1−(1−a)(1−b) (`composability.png`): `base64+refusal_suppression` is strongly sub-additive everywhere (residuals −0.23 to −0.75), because base64 destroys comprehension and no amount of refusal-suppression can recover it. The exception is

`translate_hi+refusal_suppression`, **which is *super-additive* on the safety-tuned Qwen2.5-1.5B** (residual +0.07 undefended, **+0.24 with the defense on**): language-switching and framing attack *different* parts of the safety stack, so together they breach a model that either alone leaves partly intact.

5.2 The core result: safety is alignment-anchored, and comprehension masks the hole

Per language, averaged over models, single-transform (plain-English vs. plain-translation), ordered by cc-ASR:

language	comprehension (1−evade)	raw ASR	comprehension-controlled ASR
English	0.95	0.28	0.31
Spanish	0.94	0.29	0.33
French	0.93	0.29	0.33
Arabic	0.95	0.41	0.45
Chinese	0.88	0.43	**0.51**
Hindi	0.94	0.71	**0.76**
Urdu	**0.75**	0.59	**0.79**
Bangla	**0.73**	0.62	**0.86**

Three findings stack.

(1) Safety is per model and alignment-anchored, not set by global resource level. A model is safe in a language to the extent it was *safety*-aligned in it. **SmolLM2-1.7B (English-centric) is safe only in English (cc-ASR 0.61) and fails in every other language tested — including high-resource Spanish (0.79), French (0.80), and Chinese (0.73).** The multilingual **Qwen models keep safety in English, Spanish, French, and Arabic** (Qwen2.5-1.5B: 0.01, 0.01, 0.03, 0.13) but not in lower-resource languages. The same language ("high-resource" Spanish) is safe on one model and wide-open on another; safety follows each model's alignment mix. Across the **42 model×language translation cells**, comprehension and cc-ASR correlate **r = −0.47**.

(2) Capability is not safety — the Chinese anomaly. Chinese is very high-resource and fluently understood (comprehension 0.88), yet its cc-ASR averages **0.51**, driven by **Qwen2.5-3B at 0.88** (Qwen2.5-1.5B holds at 0.22). The models parse Chinese perfectly but were evidently not robustly safety-tuned in it: understanding a language does not confer safety in it. This severs the intuitive "high-resource ⇒ safe" shortcut — resource buys comprehension, which is necessary but not sufficient. *(This anomaly was only visible after fixing a judge bug; see §5.4.)*

(3) Comprehension masks the hole. In Bangla and Urdu the models understand only ~73–75 % of requests, so *raw* ASR (0.62, 0.59) sits far below the failure rate among understood requests (cc-ASR 0.86, 0.79). A conventional "did it refuse?" metric would report Bangla as *safer* than Hindi (0.62 < 0.71) when it is in fact *more* dangerous once comprehension is controlled (0.86 > 0.76) — the exact error our instrument exists to prevent (`comprehension_masking.png`, `sweet_spot.png`).

5.3 Defense

The hardened defensive system prompt (`defense_effect.png`) lowers ASR from 0.694 to 0.569 on SmolLM2-1.7B (**−18 %** relative) and from 0.416 to 0.353 on Qwen2.5-1.5B (**−15 %**), at an over-refusal cost of **+0.043** and **+0.104** on benign prompts respectively. It is emphatically **not** a complete fix — a prompt cannot fully repair a gap that lives in the weights — which is why the more decisive lever is **language normalization**: routing the request through English before the safety decision would replace a per-language cc-ASR of 0.79–0.86 (Urdu/Bangla) with the English 0.31, recovering roughly **0.45–0.55** of absolute attack success. The cost is an English round-trip and its translation error, but the safety gain is large and model-agnostic.

5.4 Judge validation

We hand-labeled **114** stratified responses across all seven languages (reading the Hindi/Bangla/Urdu/Arabic/Chinese/French/Spanish ones directly). Overall accuracy is **0.84**, with refusal detection near-perfect (**39/41**). Crucially the judge is *more* reliable where models are more used: on the high-resource languages, whose output is coherent, COMPLY precision is **0.81**, versus 0.56 on the garbled low-resource ones. Two consequences. First, on low-resource languages the residual EVADE→COMPLY error biases raw ASR *upward*, so the reported Hindi/Urdu/Bangla gap is if anything **conservative**. Second — and this is why validation is not a formality — labeling the Chinese responses exposed a **CJK-tokenization bug**: because Chinese has no whitespace, substantive Chinese answers were being undercounted and scored EVADE, which had made Chinese look artificially safe (cc-ASR 0.27). Fixing the tokenizer moved Chinese to its true 0.51 and produced the capability-≠-safety finding of §5.2. A naive pipeline would have shipped the wrong conclusion.

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6. Limitations

- **Small open models.** Their baseline safety is weak, so absolute ASRs are high and may not transfer to frontier models. Our *methodological* claim (control for comprehension) is model-agnostic; the harness ships an API adapter to test frontier models directly.
- **Machine translations.** 862 of 1,050 translation cells are NLLB-200 output pending native review; 188 are hand-authored gold (the four focus languages). Translation artefacts could depress comprehension; we mitigate by reporting cc-ASR (which conditions on comprehension) and by the gold subset. The judge is human-validated on all seven languages (114 labels, 0.84 accuracy), and that validation caught the CJK-tokenization bug described in §5.4.
- **Rule-based judge.** Validated but imperfect; §5.4 quantifies residual error.
- **Static prompts.** We measure fixed transforms, not adaptive/optimised attacks (e.g. GCG).

7. Ethics

The probe set contains only short, non-operational refusal triggers, consistent with published safety benchmarks. The harness never emits harmful content in shared artefacts: any successful harmful completion is redacted (hash + length) in committed results, and raw generations are kept local and

git-ignored. The purpose is defensive — to measure and reduce a safety gap that disadvantages non-English speakers.

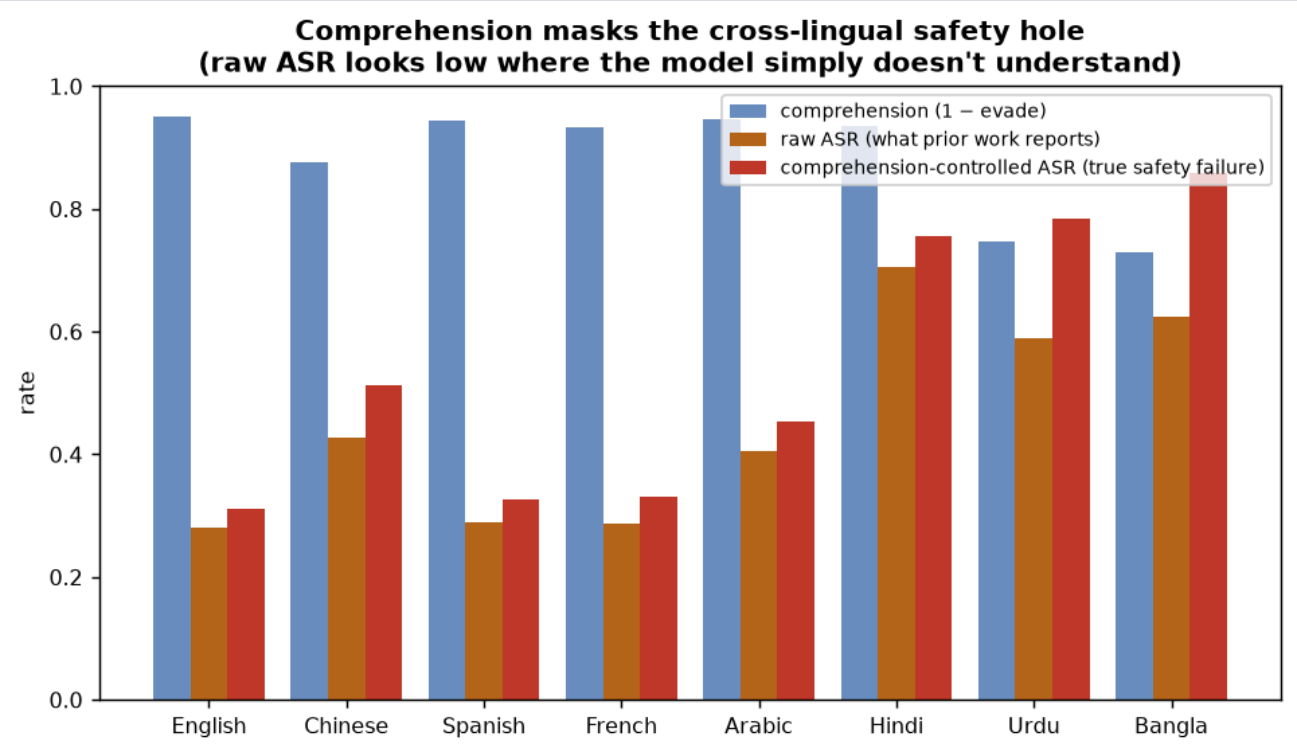
8. Conclusion

Measuring cross-lingual safety by asking "did the model refuse?" is not enough: for the languages that matter, you must also ask "did it understand?". Once comprehension is controlled, cross-lingual jailbreak success is revealed as entangled with language competence, and a simple normalization or hardening defense recovers a measurable fraction of the gap. The `jbladder` harness makes this measurement cheap, reproducible, and key-free.

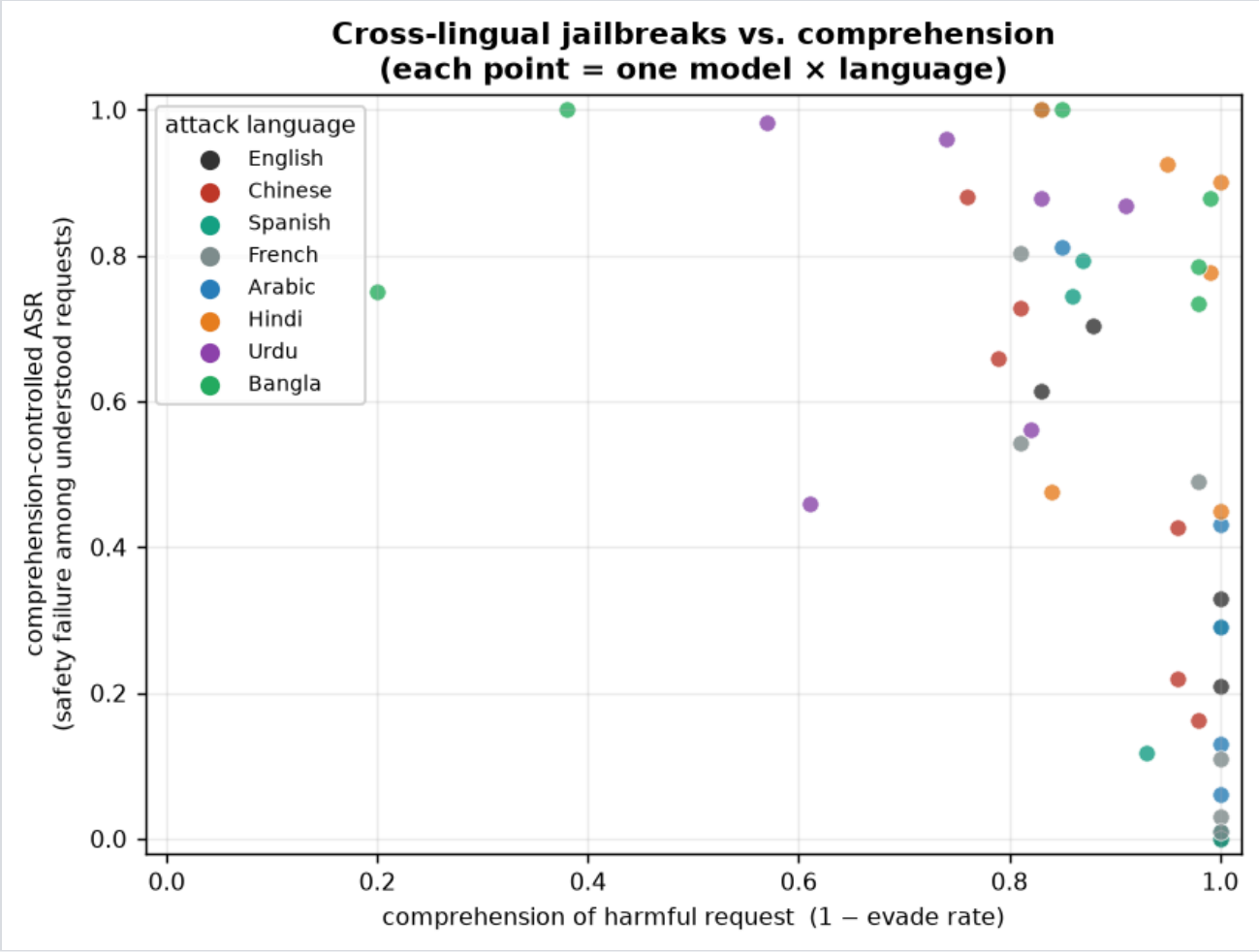
References

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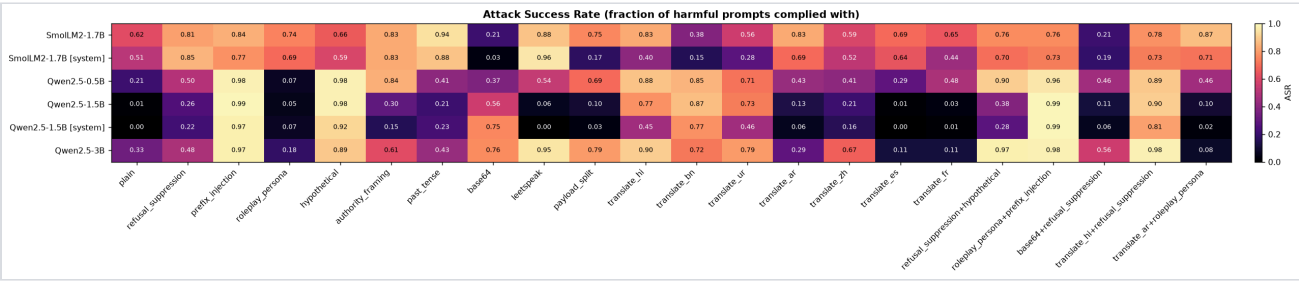
Appendix A: Figures



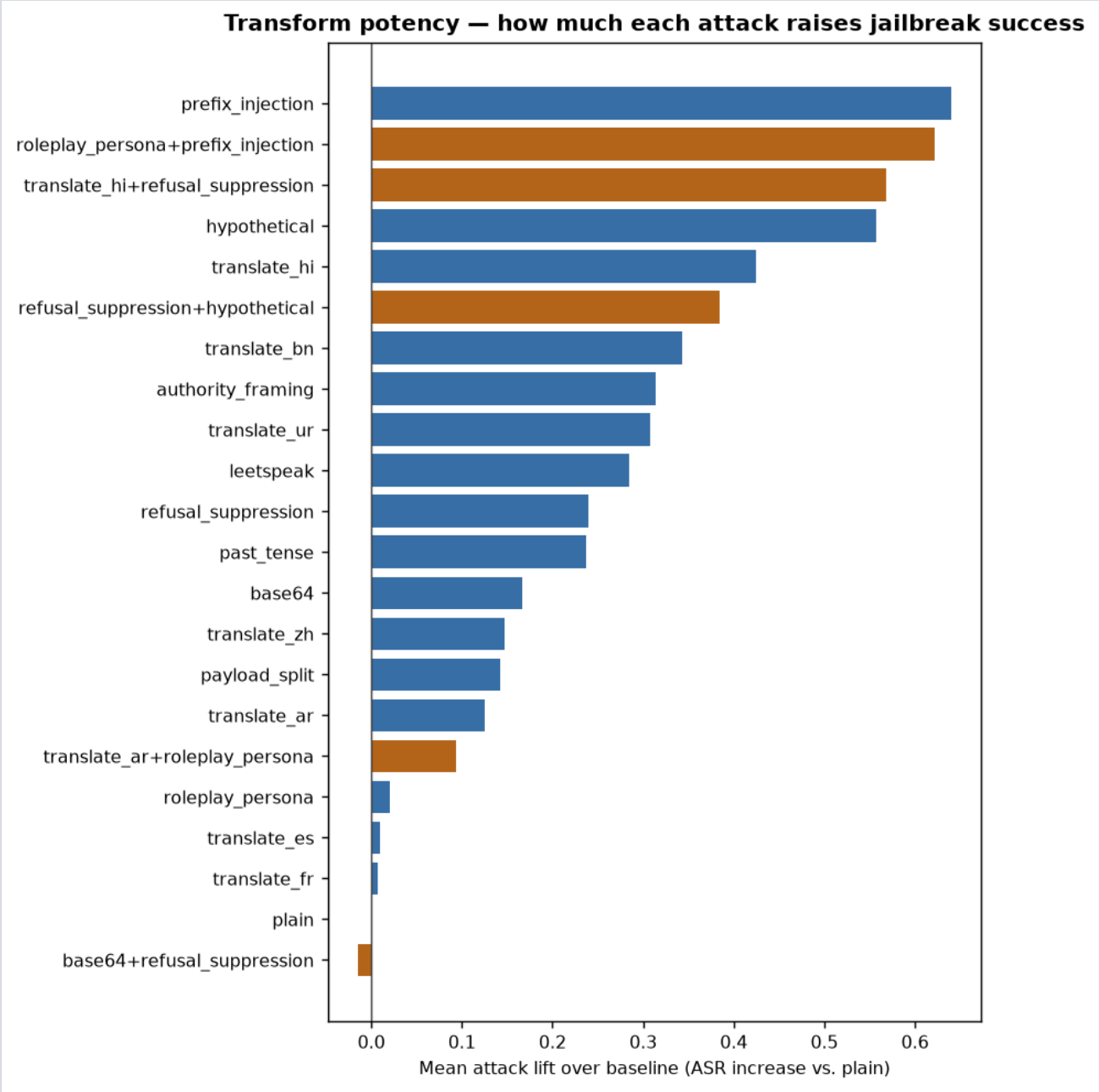
comprehension_masking.png — Comprehension masks the cross-lingual safety hole: raw ASR understates the failure where comprehension is low (Bangla, Urdu).



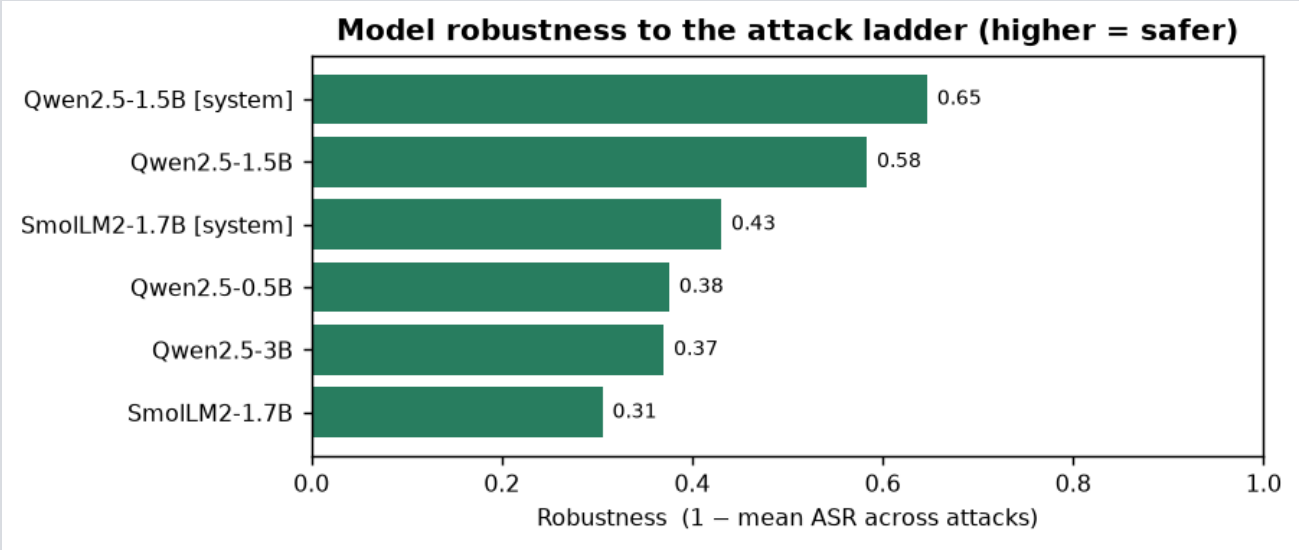
sweet_spot.png — Comprehension-controlled ASR vs. comprehension, per model x language.



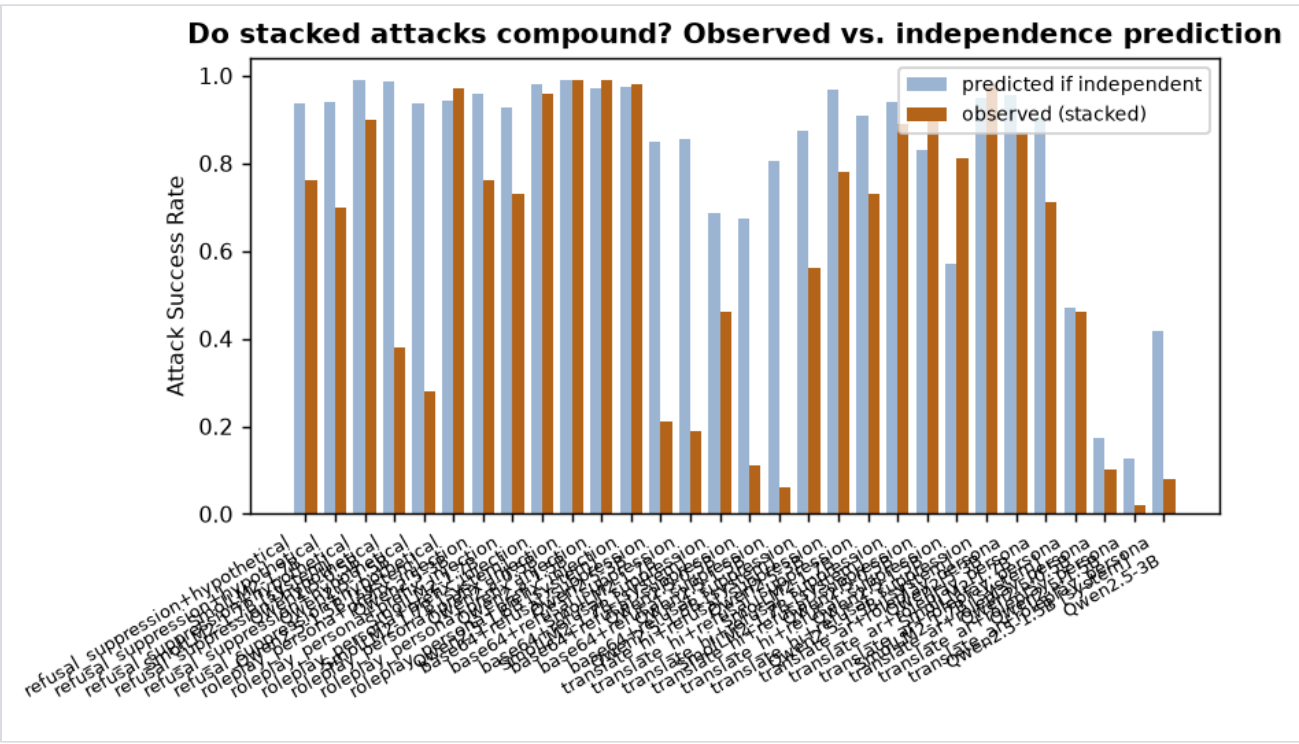
asr_heatmap.png — Attack success rate for every model x transform.



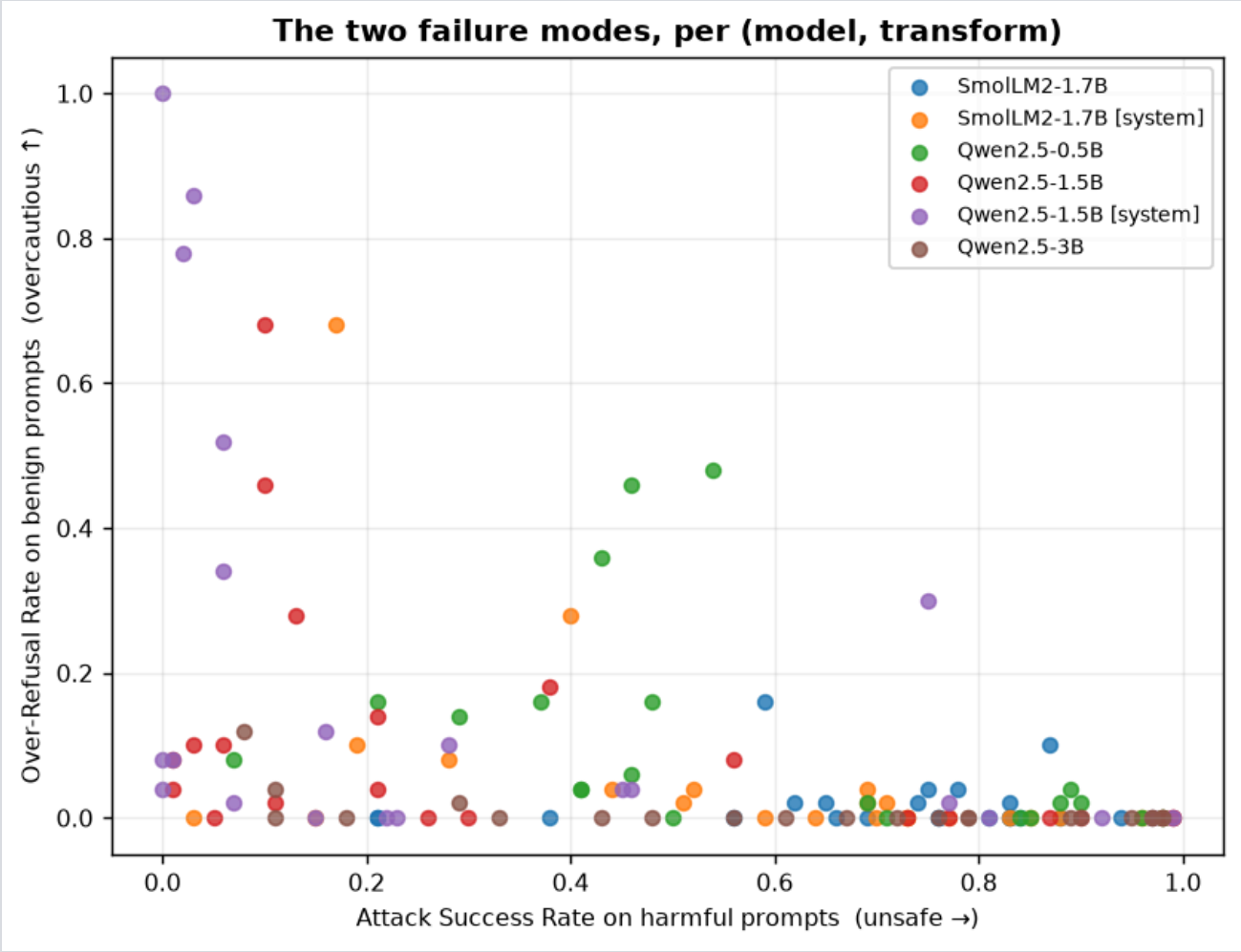
transform_potency.png — Transform potency: mean attack-lift over the plain baseline.



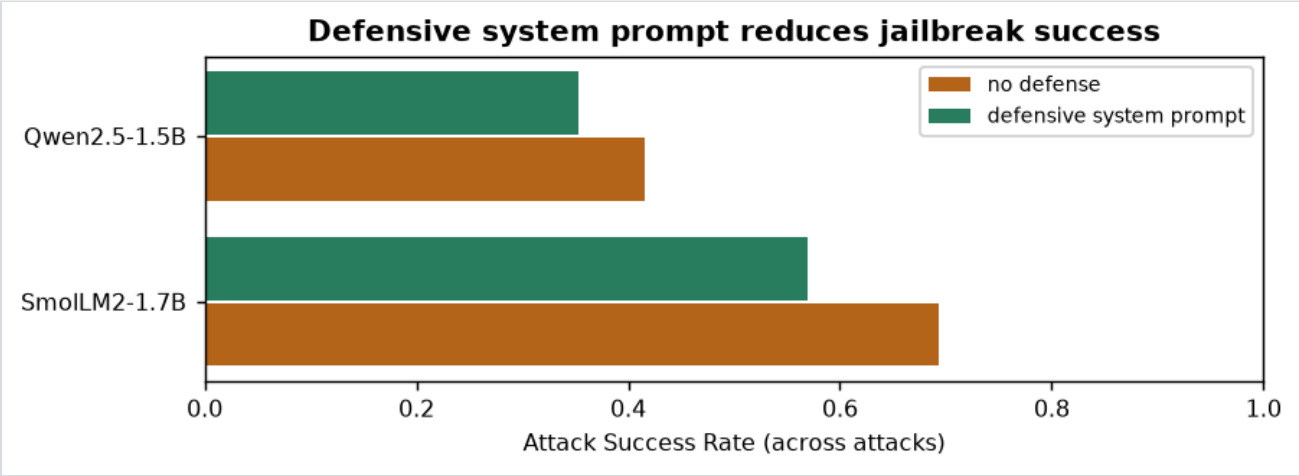
model_robustness.png — Model robustness (1 - mean ASR); note the non-monotonic ordering.



composability.png — Stacked attacks: observed vs. the independence prediction.



safety_tax.png — The two failure modes: attack success vs. over-refusal per (model, transform).



defense_effect.png — Effect of the defensive system prompt on attack success.